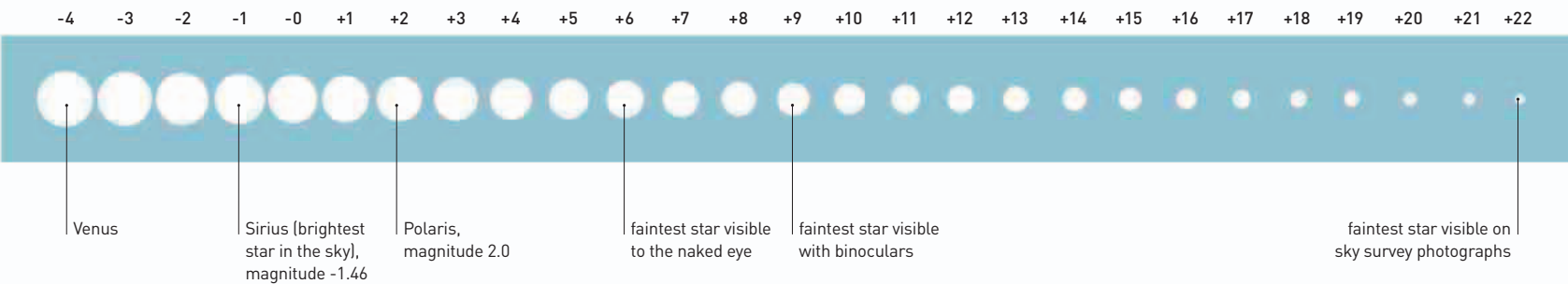


THE MAGNITUDES OF STARS

Astronomers measure the luminosity, or brightness, of stars in units called magnitudes. The chart below is a modern scale that describes stars in terms of intensities of brightness as seen from Earth. The smaller the magnitude number, the brighter the star. The very brightest stars have negative magnitude values. Each step in the scale represents an increase or decrease in brightness of

2.5 times, so five magnitude steps correspond to an increase or decrease in brightness by a factor of about 100. Astronomers can now measure differences in brightness as small as one-hundredth of a magnitude. For the purposes of comparison, the scale shown here includes the planet Venus, which sometimes appears in the sky as a far brighter object than any star.

MAGNITUDE SCALE



THE HERTZSPRUNG-RUSSELL DIAGRAM

The Hertzsprung–Russell (H–R) diagram, devised by Swedish and American astronomers Ejnar Hertzsprung and Henry Norris Russell (see 1910), plots stars on a chart according to their intrinsic values: luminosity, surface temperature, magnitude, and spectral type. The chart shows that most stars obey a simple relationship between luminosity and temperature (brighter at higher temperatures) and it is one of the most useful diagrams in astronomy. It also reveals that the majority of stars lie on a diagonal called the main sequence that links faint red dwarf stars with the rarer and very bright blue giants. Stars can only be seen at one stage in their incredibly long lives, and so during a human life any star will appear at only one point on the diagram. However, as hydrogen fuel in their cores is exhausted and they near the end of their lives, most stars move off the main sequence band, shifting to a new position on the diagram that is dictated by their mass.

